Vasudeva Kumar Mantravadi

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SUMMARY

- Visa Status: Open to Relocate | Nationality: Indian | Age: 48
- Accomplished Chemical Engineer and R&D Specialist with over 20 years of experience in Process Engineering, specializing in Pre-FEED, FEED, and Detailed Engineering for Onshore and Offshore projects. Expertise includes OLGA, LedaFlow, K-Spice, and ASPEN HYSYS dynamic simulation, real-time monitoring, and production optimization, delivering multimillion-dollar impacts across upstream assets and FPSOs. Deeply familiar with integrated UAE permit-to-work systems and global industry standards for oil and gas operations.

EXPERIENCE

PRINCIPAL PROCESS ENGINEER

FALKOR (Previously KONGSBERG DIGITAL INDIA SOFTWARE AND SOLUTIONS PVT LTD.) **April 2020 - Present, Bangalore, India**

- Client / Project: ONGC, Reliance, Snohvit Future Phase 2
- Led the end-to-end development of front-end, engineering, real-time, and Operator Training Simulator (OTS) solutions for upstream oil and gas customers.
- Created an AI agent called K-Spice Tutor that improved training efficiency for process engineers by 40% through guidance and workflow support.
- Designed and implemented a standalone SRP calculation engine using GenAI for diagnostic and calculation tools, integrated into a real-time simulation and visualization platform.
- Enabled data exchange from an end-to-end LedaFlow-K-Spice dynamic simulator, performing SRP calculations and streaming results to the visualization layer in real time.
- Architected and deployed a real-time production system for ONGC Ankleshwar and delivered a Production Management System (PMS) for the ONGC Nandasan onshore field.
- Developed and delivered Production System Simulators for Reliance KGD6 R-Cluster, SAT-Cluster, and MJ offshore fields (FPSO), providing critical start-up and monitoring support.
- Conducted FEED/EPC studies for verification of design based on developed simulators, providing critical input to modify/upgrade design for new compressor installations at the Snohvit Future Phase 2 project.
- Led the winning team at #AIHackathon 2025, developing a #GenAI and #LLM-powered hybrid gas lift optimization solution for mature wells.
- Directed comprehensive HAZOP analysis for major oil and gas production assets, ensuring compliance with ISO 17776 and improving incident response time by 35%.
- Directed full lifecycle Project Management for five AI-driven simulation and production system deployments over 24 months, leveraging agile methodologies and tools to deliver all milestones on schedule while boosting operational KPIs by up to 30% across multiple upstream assets.

PROCESS ENGINEERING SPECIALIST

SHELL TECHNOLOGY CENTER BANGALORE **March 2015 - March 2020, Bangalore, India**

- Client / Project: Shell chemicals and refining sites
- Managed a multi-year, multi-million dollar Upstream Simulation Tools project, collaborating with Flow Assurance, SLB development, and Process Engineering teams.
- Deployed enterprise simulation tools for asset optimization across a global upstream and downstream portfolio, enabling improved production and margin decisions on multi-billion-dollar assets.
- Created simulations for DCS control schemes, leading to a 20% decrease in commissioning time and enhanced process stability.
- Led the global deployment of AspenOne Engineering Suite across all Shell chemicals and refining sites, including training for 30 engineers in Singapore.
- Led cross-functional teams to ensure timely delivery of engineering tools with a strong emphasis on quality and process safety.
- Provided customized training and mentored graduates to promote technical excellence and diversity.
- Directed the implementation of Process Engineering solutions for asset integrity projects at 18 global sites over 5 years, deploying AspenOne and OLGA tools to enhance plant performance by 15% and achieve strict process safety KPIs.

Scientist-Control & Optimization

ABB GLOBAL INDUSTRIES AND SERVICES LIMITED **September 2011 - March 2015, Bangalore, India**

- Client / Project: RO Desalination Plants (Jeddah, Saudi Arabia & Gujarat, India)
- Spearheaded performance monitoring for upstream oil and gas assets, conducting techno-commercial analysis and developing a Unisim prototype.
- Led design optimization studies for solar desalination processes integrated with renewable energy, identifying $50 million annual profit potential.
- Executed a pilot for an optimization solution at a large RO desalination plant in Jeddah, delivering a 2% productivity improvement and $450K in annual benefits.
- Validated energy consumption reduction of 2% at an RO desalination plant in Gujarat, resulting in estimated $100K annual savings.
- Identified energy efficiency opportunities at a major desalination plant in the UAE, projecting $200K annual benefit through specialized auditing tools.
- Authored three patent filings in renewable energy integration and model-based optimization for irrigation canal operations.

Scientist/Engineer

PROCTER & GAMBLE **January 2010 - August 2011, Bangalore**

- Client / Project: Global Manufacturing Sites
- Created and implemented a flow model library for process engineers at multiple global plants, showing potential for $100 million in yearly savings.
- Engineered a pneumatic conveying modeling platform for solid and non-Newtonian fluids, improving assessment accuracy by 35%.
- Managed end-to-end process design optimization initiatives for global manufacturing sites, utilizing ASPEN HYSYS and IDEAS to reduce costs by 22%.
- Troubleshot and resolved critical scrubber blow-down issues during plant shutdowns to ensure process safety and operational continuity.
- Conducted steady-state and dynamic simulations using ASPEN HYSYS to enhance chemical process designs.
- Validated and deployed engineering tools globally by conducting comprehensive training sessions for engineering teams.
- Directed Process Engineering initiatives across 12+ global manufacturing sites over 3 years, leveraging ASPEN HYSYS and IDEAS to optimize production throughput by 28% and surpass targeted reliability KPIs by 19%.

Process Simulation Engineer-II

INVENSYS INDIA DEVELOPMENT CENTER PVT. LTD **February 2009 - January 2010, Hyderabad**

- Client / Project: ExxonMobil
- Developed valve flashing calculation features and addressed critical software issues for DySim simulation software, improving result accuracy by 95%.
- Resolved 15 technical challenges in collaboration with ExxonMobil, improving system efficiency.
- Added 5 feature enhancements to improve software reliability and performance based on global client feedback.
- Facilitated knowledge transfer by preparing detailed user guides and technical documentation for DySim enhancements to streamline onboarding for new users and support teams.
- Conducted validation testing of DySim's simulation results against actual plant data, identifying model improvement opportunities and increasing stakeholder confidence in simulation accuracy.

EDUCATION

Postdoctoral Fellow in Chemical Engineering

MAX PLANCK INSTITUTE FOR DYNAMICS OF COMPLEX TECHNICAL SYSTEMS (MPI) • Magdeburg, Deutschland • 2009

- Worked on Kinetics of Iron oxide for purification of Hydrogen for Fuel cell applications
- Experimental investigations on in-house reactor to determine the kinetics for reduction reaction with H_2 and oxidation with H_2O.

Doctor of Philosophy in Chemical Engineering

INDIAN INSTITUTE OF TECHNOLOGY, Madras • Chennai • 2009

- Thesis is on Modelling and Nonlinear Analysis of Chemical and Biological Reactor-Separator Systems
- Significant output includes seven International Journal and Conference Publications.
- Secured 9.35(/10.0) CGPA.

Master of Technology in Chemical Engineering

REGIONAL ENGINEERING COLLEGE • Warangal • 2002

- Specialization in Computer Aided Process Equipment Design.
- First in the batch with an aggregate percentage of 78.4.
- Master's thesis on Modelling and Simulation of a bioreactor for ethanol synthesis using Artificial Neural Network.

Bachelor of Technology in Chemical Engineering

ACHARYA NAGARJUNA UNIVERSITY • Guntur • 2000

- Second in the University in Chemical Engineering discipline with an aggregate percentage of 80.01.

CERTIFICATIONS

Basic Offshore Safety Induction and Emergency Training (BOSIET) with Emergency Breathing System Certification

WOISO • 2022

SKILLS

Engineering Software: K-Spice, LedaFlow, OLGA, Aspen Plus, gPROMS, ASPEN HYSYS, Unisim, DySim, Extend, IDEAS, Visual Studio, Generative AI, LLM

Field Competencies: Pre-FEED, FEED, Detailed Engineering, Dynamic Simulation, Transient Simulation, Production Optimization, Real-Time Monitoring, Compressor Anti-Surge, OTS / DCS Control Testing, Hydraulic Calculations, HAZOP, Asset Integrity, Energy Efficiency, Desalination Engineering

Management: Technical Leadership, Cross-Functional Collaboration, Innovation Management, Training & Mentorship